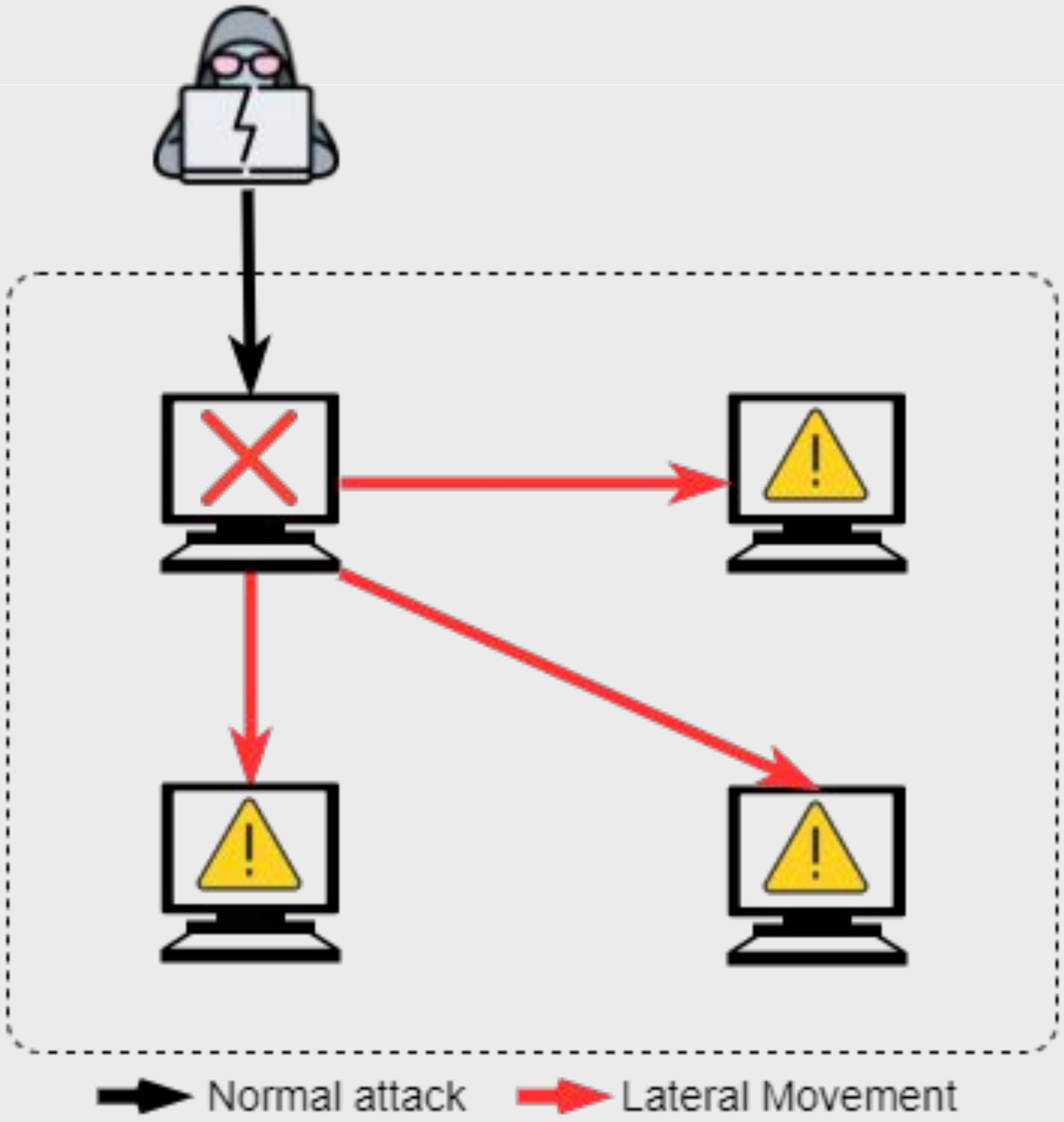
Prediction System of Lateral Movement from Sysmon based on ATT&CK

for Blackhat Asia 2023

Preliminary

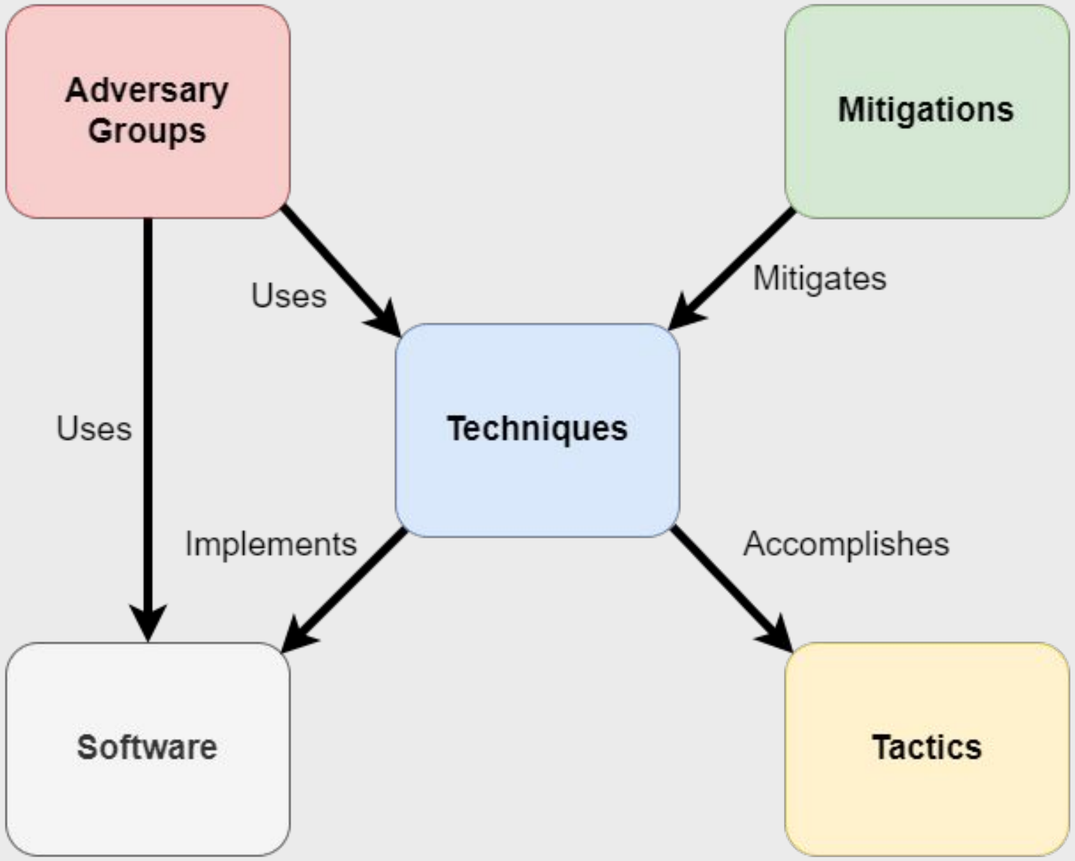
Lateral Movement

- One of the attacks of the infection expansion phase of the cyberattack.
- After hacking into the system, the attacker tries to break into other devices of the same network.
- Attackers get credential information by expanding the infection.



MITRE ATT&CK

- A knowledge base of cyber attack tactics and techniques based on past cyberattacks.
  - **Adversary Groups**: Groups of attackers
  - **Tactics**: Objective of attack
  - **Techniques**: Technique used in attacks
  - **Software**: Tools used in attacks
  - **Mitigations**: Mitigations for Attacks



Atomic Red Team

- An open-source library of test based on MITER ATT&CK.
- You can use it to simulate adversarial activity in their environments.

MITRE ATT&CK Technique

Adversaries may transfer tools or other files from an external system into a compromised environment. Tools or files may be copied from an external adversary-controlled system to the victim network through the command and control channel or through alternate protocols such as ftp. Once ...

Embodiment

Atomic Red Team

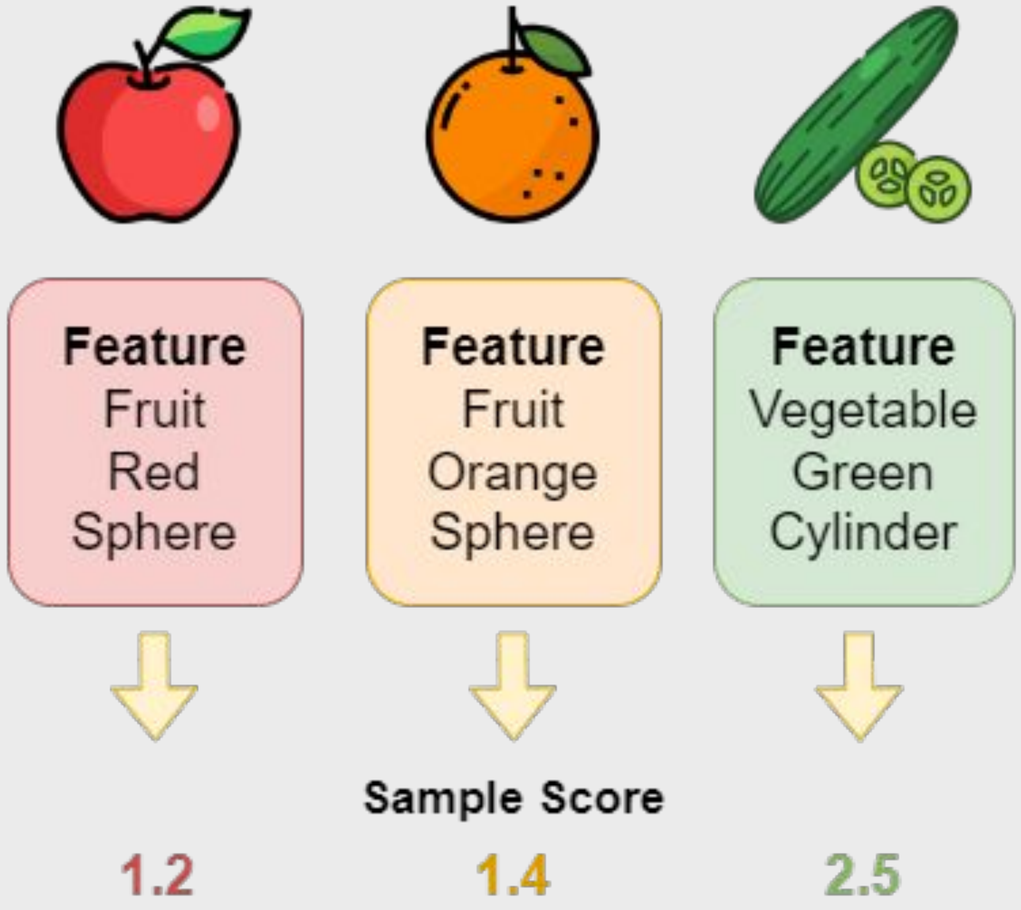
```
$ rsync -r #{local_path} #{username}@#{remote_host}:#{remote_path}

$ scp #{local_file} #{username}@#{remote_host}:#{remote_path}

$ sftp #{username}@#{remote_host}:#{remote_path} <<< $'put #{local_file}'
```

The Quantification Theory Type 3

- One of the multivariate analysis.
- Sample score are obtained by calculating what elements are in each sample.
- The quantification theory type 3 analyzes the similarity based on the the proximity of sample score.



Sysmon

- Tool for recording Windows system activity.
- We can investigate the cause and behavior of the system.


```
<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'>
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<Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' />
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<Level>4</Level>
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</System>
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<DataName='ProcessGuid'>{a188cd34-5516-6307-6c06-000000000400}</Data>
<DataName='ProcessId'>16764</Data>
<DataName='Image'>C:\Windows\System32\SearchProtocolHost.exe</Data>
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<DataName='Description'>Microsoft Windows Search Protocol Host</Data>
<DataName='Product'>Windows® Search</Data>
<DataName='Company'>Microsoft Corporation</Data>
<DataName='OriginalFileName'>SearchProtocolHost.exe</Data>
<DataName='CommandLine'>"C:\windows\system32\SearchProtocolHost.exe"
Global\UsGthrFltPipeMssGthrPipe14_Global\UsGthrCtrlFltPipeMssGthrPipe14 1 -2147483646
"Software\Microsoft\Windows Search" "Mozilla/4.0(compatible; MSIE 6.0; Windows NT; MS
Search 4.0 Robot)""C:\ProgramData\Microsoft\Search\Data\Temp\usgthrsvc" "DownLevelDaemon"
</Data>
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/Data>
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<DataName='ParentCommandLine'>C:\windows\system32\SearchIndexer.exe /Embedding</Data>
<DataName='ParentUser'>NT AUTHORITY\SYSTEM</Data>
</EventData>
</Event>
```

About Our Tool

Functions

Conversion of Sysmon logs

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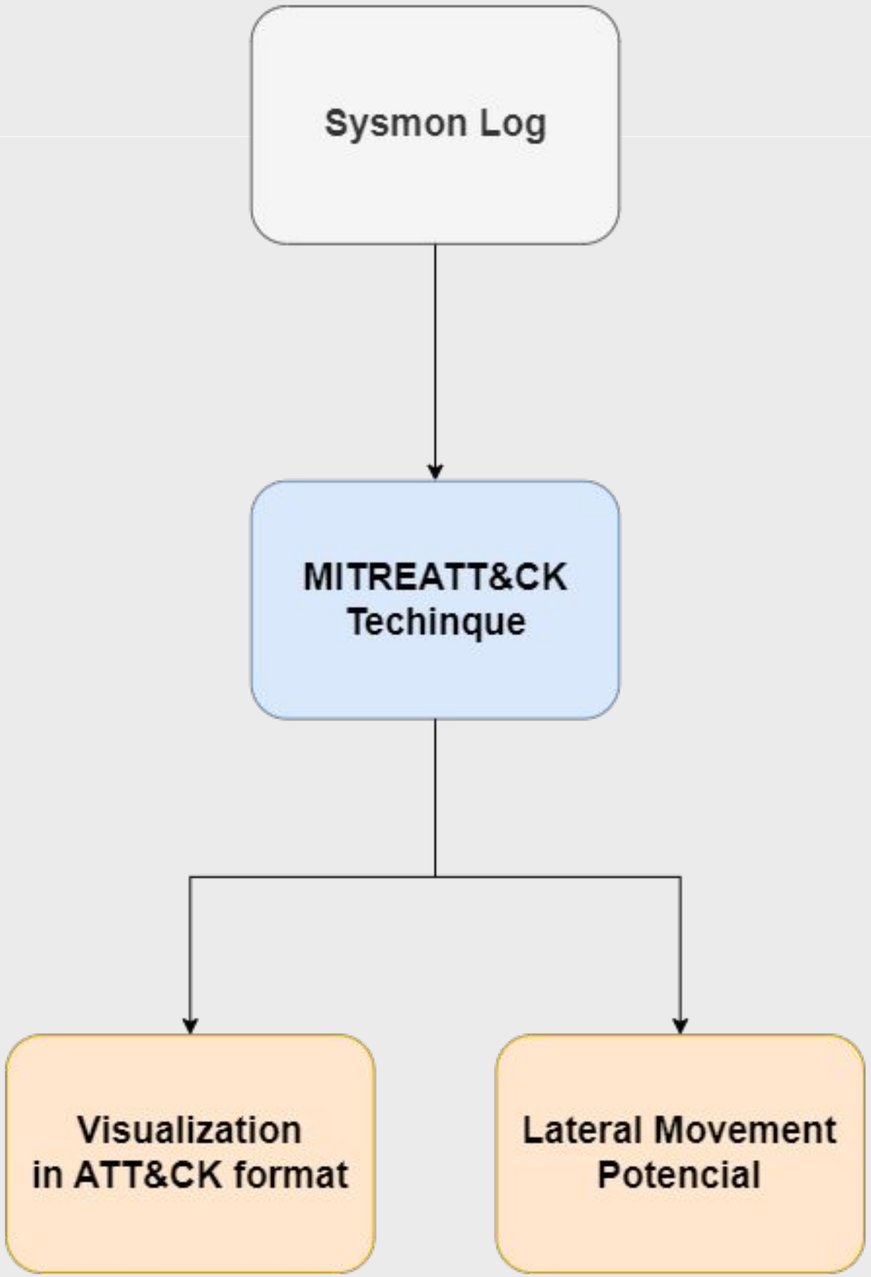
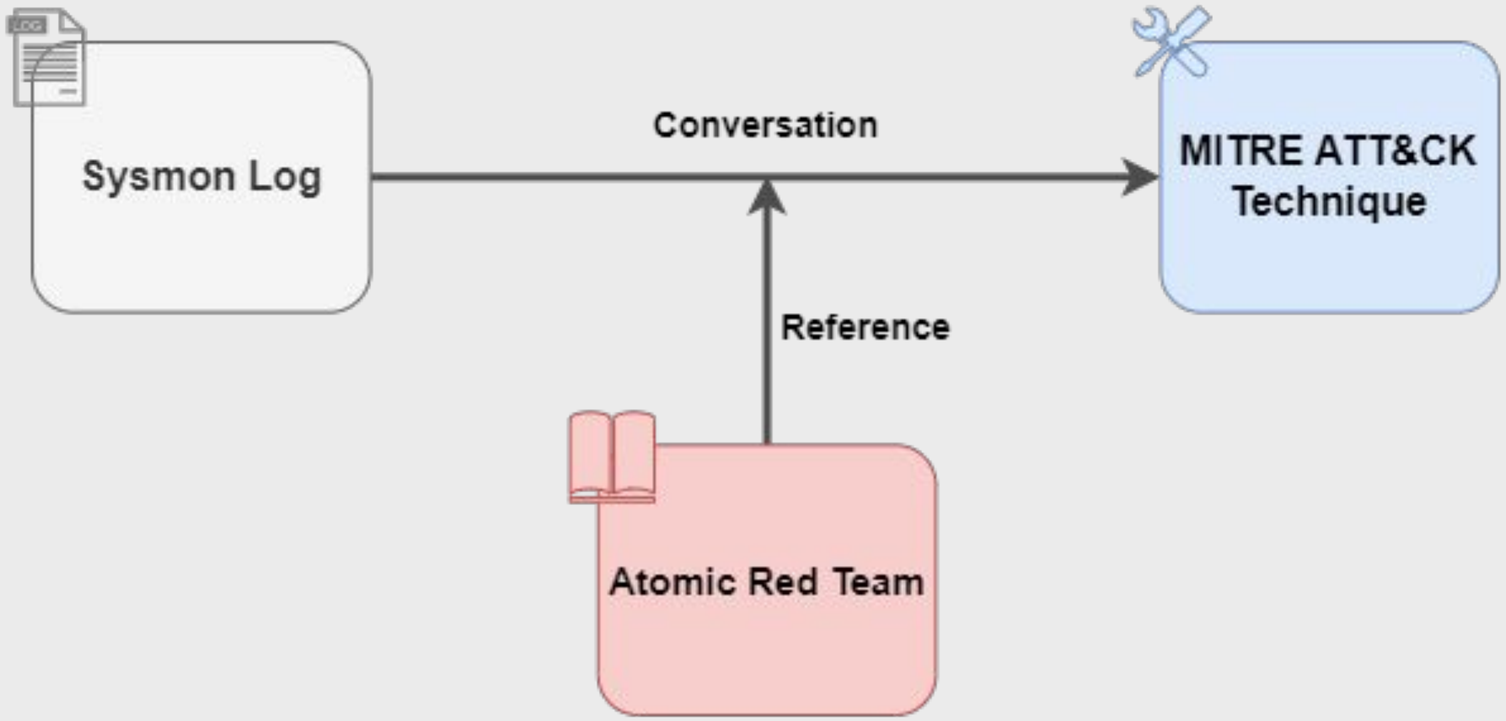
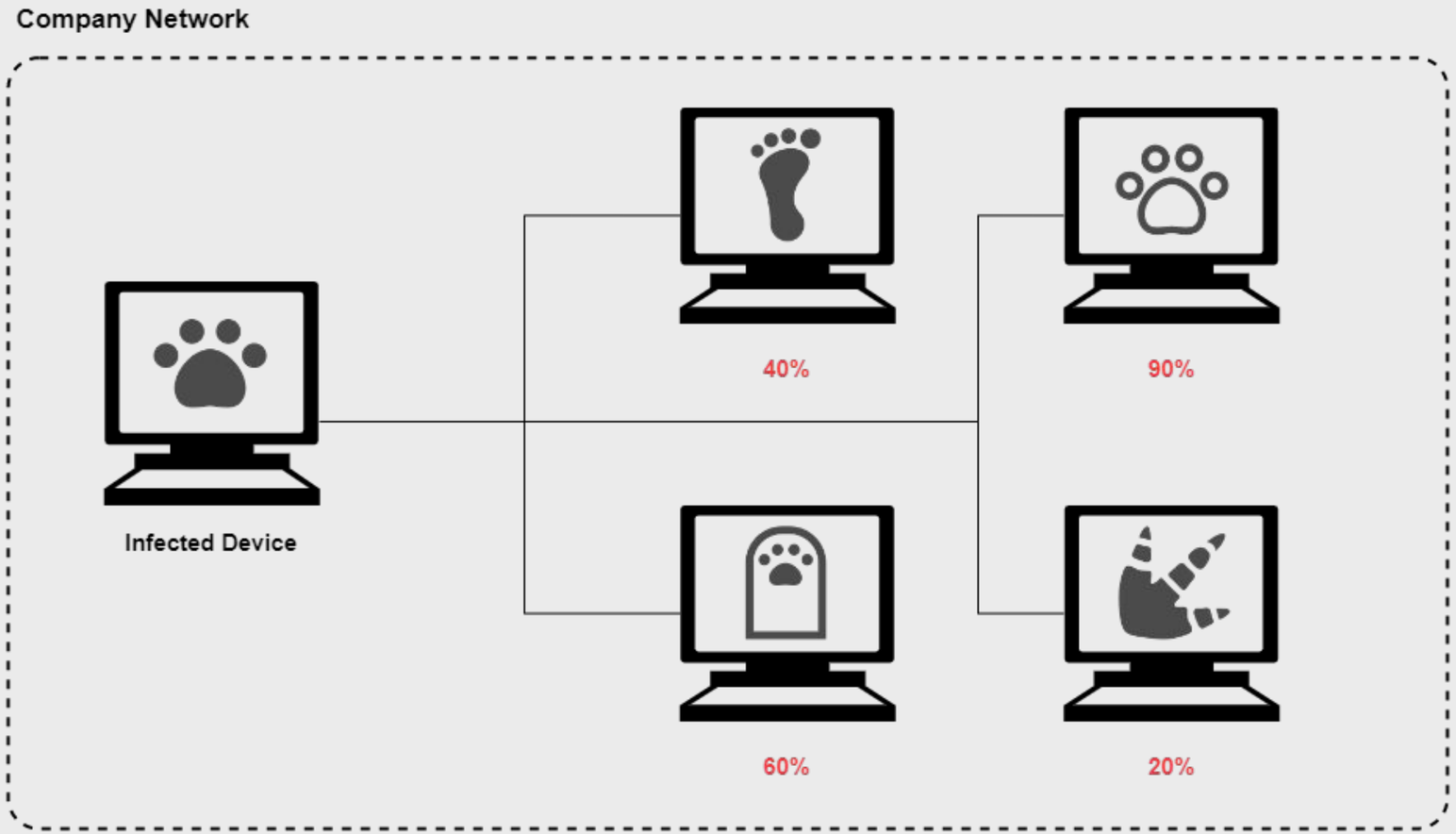
Prediction of Lateral Movement Potencial



Conversion of Sysmon logs

- This tool converts Sysmon logs to ATT&CK Techniques.
  - Reference to Atomic Red Team.
- The attacks are displayed as a list or colored ATT&CK Matrix.



Prediction of Lateral Movement Potencial

- This tool can find the device in Lateral Movement from the infected device's log similarity to others.